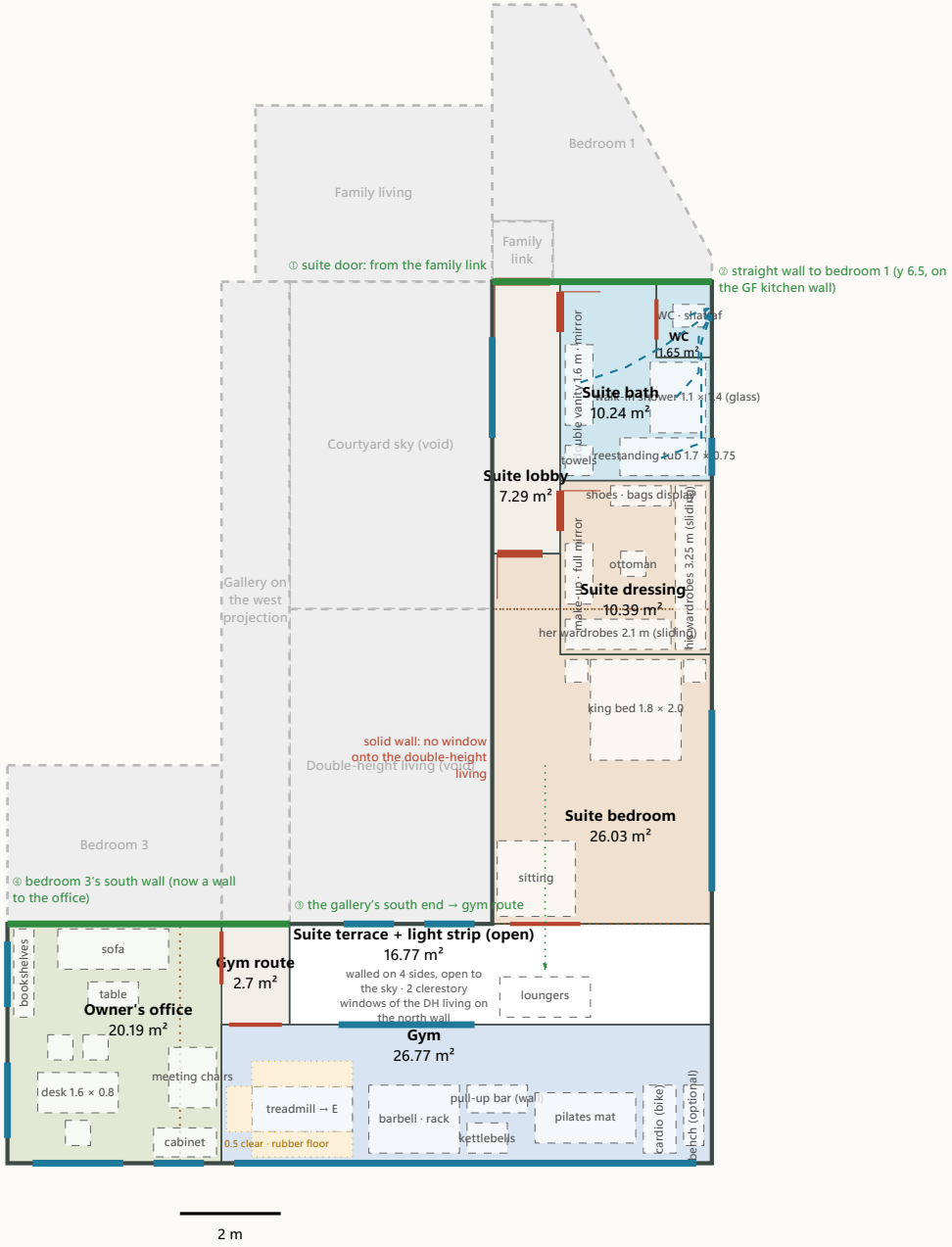
First floor · the L: owner's office → gym route → gym → court → master suite

(x east, y south) · wall centre lines · north up

First floor V5.1 · part 2 of 2 (part 1: studies/ff_family.typ) · owner's office, gym (treadmill at the west end), suite court and master suite



Rooms

Room	Axes w × d (m)	Axes m²	≈ clear m²
Suite lobby	1.35 × 5.4	7.29	5.94
Suite bath	polygon	10.24	8.79
Suite WC (closet)	1.1 × 1.5	1.65	1.09
Suite dressing	3.01 × 3.45	10.39	9.01
Suite bedroom	polygon	26.03	23.56
Suite terrace + light strip (open)	8.39 × 2	16.77	–
Gym route	1.35 × 2	2.7	2.03
Owner's office	4.25 × 4.75	20.19	18.16
Gym	9.74 × 2.75	26.77	23.96
Zone		**122.04**	

Clear ≈ axes minus half a wall on every edge (250 mm outer, 200 mm inner, nothing on open edges), corners ignored.

✓ All rooms inside the zone · no overlaps · no unassigned area · furniture inside its room, not overlapping, out of the lobby route and the gym walkway · treadmill 0.5 m behind and each side, gym door free · door swings clear · every room has a door reached from the lobby (or its room) · doors from the family part on the interfaces · lobby ≥ 1.0 m clear · no window onto the double-height living

Key numbers

- **Suite:** lobby 7.29 · bath 10.24 + WC closet 1.65 · dressing 10.39 · bedroom 26.03 = **55.6 m² indoors** + terrace 16.77 m².
- **Gym** 26.77 m² · **suite court (terrace + light strip)** 16.77 m² · **gym route** 2.7 m² · **owner's office** 20.19 m².
- **Walked:** suite door → bedroom door ≈ 5.4 m · bedroom door → bath door ≈ 5.5 m · bed → terrace ≈ 3.9 m.
- **Interfaces with the family part (green, fixed):** ① suite door at the end of the family link; ② the straight wall to bedroom 1 on y 6.5; ③ the gallery's south end, where the gym route starts; ④ bedroom 3's south wall (now a wall to the office).

Plan: red = doors (thin = sliding / no leaf) with swing line · blue = windows · blue dashed = drains · green = interfaces · brown dotted = ground floor below · grey = the family part (fixed) · dashed boxes = furniture and fixtures.

Master suite + gym + suite court · rooms, drains, what changed, questions · V5.1

Suite lobby · 1.35 × 5.4 m

- The suite door (from the family link) opens into a 1.35 m distributor (≈ 1.15 m clear; was 1.6, the 0.25 m went to the bath and dressing): bath, dressing and bedroom each open off it; nobody walks through the dressing. A window onto the courtyard on its west wall.
- **Bent entry:** the bedroom door is at the lobby's south end, round the corner of the dressing: the bed is not seen from the suite door.

Suite bath (luxury, compact) · 3.01 × 3.95 m incl. its WC closet, 11.89 m²

- **Freestanding tub** 1.7 × 0.75 m along the south wall, by an east window; **walk-in glass shower** 1.1 × 1.4 m on the east wall; **double vanity** 1.6 m on the west (lobby) wall, mirror on that wall, facing the east window light; **towel cupboard**; open floor in the middle.
- **Door** from the lobby at the north end of the west wall (y 6.7–7.5), opening in, swing clear; the north wall stays clear as the path from the door to the WC closet, so nobody at a basin blocks the door or the WC.
- **WC in its own closet** (1.1 × 1.5 m, ≈ 0.9 m clear, sliding door) in the north-east corner, at the drain stack, with a shattaf.
- **Owner's decision (19 Oct 2026):** keep the bathtub **and** the separate shower, two basins, the WC in its own closet.
- Door from the lobby (swing clear). Squeezed from 13.55 to 11.89 m² keeping every luxury item (tub, walk-in shower, two basins, WC closet); the 0.55 m went to the dressing.

Suite dressing (luxury walk-in) · 3.01 × 3.45 m

- **Door** from the lobby at the north end of its west wall (y 10.65–11.45), like the bath door; opening in, swing clear.
- **His wardrobes** 3.25 m along the east wall and **her wardrobes** 2.1 m along the south wall, both with **sliding doors** (no swing space); a **shoe and bag display** 1.2 m on the north wall; a **make-up table with a full-length mirror** on the west wall; a small **ottoman** in the middle. 5.35 m of wardrobes in all (was 4.7). No window: good lighting inside the wardrobes and over the mirror.

Suite bedroom · 26.03 m²

- **King bed** 1.8 × 2.0 on the dressing wall, east of the door line, **facing south to the terrace and the palms**; bedside tables. A sitting corner on the solid west wall.
- **West wall solid:** the double-height living is behind it, no window (checked). Windows east and the sliding door to the terrace (south).

Suite court (terrace + light strip) · one private court, 8.39 × 2.0 m (V5.0 7.71), walled on four sides, open to the sky

- **One space:** the old light strip and the suite terrace together, 16.77 m², from the bedroom only (sliding door). No railing between them any more; the railing / parapet question is now only on the gym edge (y 21.25).
- **The clerestory windows** of the double-height living (two 1.0 × 1.0 m, in its south wall y 19.25) now open onto the suite terrace: their sill must stay above eye level from the terrace floor (or obscured glass), so the terrace does not look down into the living. To check against the floor levels.
- **Walled on all four sides, open only to the sky:** north the double-height living's south wall (clerestory) and the suite bedroom (sliding door), south the gym's north wall, west a full wall to the gym route, east the outer wall. Fully private; it works as a light well for the clerestory windows.
- **Open to the sky:** its own floor drain and waterproofing on the men's majlis roof slab; summer heat can collect in a 2.0 m deep court, so the clerestory windows and the bedroom door should open for air.

Gym · 9.74 × 2.75 m, 26.77 m² · a rectangle (V5.1)

- **One straight bar** along the south facade (x 3.575–13.311; 0.675 m longer than V5.0, its west wall follows the office), sliding door from the gym route at its west end; it does **not** touch the suite bedroom (the suite court lies between). Windows south (long) and high windows north onto the suite court (obscured, the court is private). Engineer: loads and vibration over the men's majlis.
- **Treadmill at the west end** (V5.1), away from the master suite: 0.9 × 2.0 m along the gym's length, the user facing east, **0.5 m clear behind** (to the office wall) **and on each side**; a **dense rubber floor** (20–40 mm) under it and its

Drains (blue dashed) · no east duct any more

- **Suite bath:** all fixtures drain east in a **sunken bath floor** to one **stack on the outside face of the east wall** (x 13.311, y ≈ 6.7–7.1), down to the east-yard manhole where the kitchen already drains. WC ≈ 0.3 m (beside the stack), shower ≈ 1.5 m, vanity ≈ 2.8 m, tub ≈ 3.2 m along the east wall.
- The bath stands over the GF kitchen: **nothing hangs in the kitchen ceiling** (sunken floor), and the stack is outside the wall, clear of the hood (**hood position on the east wall to be fixed south of y 7.5**).
- No drain over the master bedroom (rule kept). The terrace drains to the east (roof falls).

Walls (engineer)

- **On GF lines:** y 6.5 (bedroom 1 / suite: the GF kitchen's north wall), the outer walls, x 8.95 (courtyard / double-height living). The suite bedroom spans the GF kitchen / living wall (y 13.0, dotted) with no wall above it.
- **Off the GF lines:** x 10.3 (lobby / bath and dressing), y 10.45 (bath / dressing), x 12.211 and y 8.0 (WC closet), y 11.9 and y 13.9 (bedroom), all over the GF kitchen; on the men's majlis roof: y 21.25 (gym / suite court), x 3.575 (office / gym route and gym), x 4.925 y 19.25–21.25 (gym route / suite court). **V5.1:** the office's west wall (x −0.675, y 19.25–24) and the south wall's west end (x −0.675–0) stand on the **outward cantilever beams**, over the GF foyer and the guest entrance.

What changed from V4.1

- The suite now starts at the **straight wall on y 6.5**: the lobby is a clean 1.6 × 5.4 rectangle, the bath takes bedroom 1's former tail and bath strip.
- **East duct removed:** the bath drains to a stack outside the east wall instead.
- Dressing slightly larger (3.5 m deep, was 2.9); bedroom and terrace unchanged.
- **Gym made L-shaped all the way east (18 Oct, undone on 20 Oct)** (10.56 × 2.75 along the south + a 1.6 × 2.0 leg up the east wall to the suite bedroom, no door to it); the terrace (20.7) moves north and joins the light strip as one open space of 16.77 m².
- **West projection (family study, 19 Oct):** the gallery runs on a 1.35 m projection along the courtyard and the double-height void (x 4.25–5.6); the gym route moves to its end, so the court loses its west 1.35 m (now x 5.6–11.711) and the gym door moves to x 4.45–5.35. The two clerestory windows move into the narrower void (x 6.0–7.0 and 7.6–8.6).
- **Luxury bath (19 Oct):** the lobby narrows to 1.35 m (≈ 1.15 clear); the bath becomes a compact luxury bath with a WC closet, walk-in glass shower, freestanding tub and a 1.6 m double vanity; dressing and bedroom follow.
- **V5.1 · split projection (24 Oct, variant SP accepted):** the west band moves 0.675 m west: the **office** (same 4.25 × 4.75 m) is now x −0.675–3.575, its west wall on the outward cantilever, a canopy over the guest entrance; the **gym route** x 3.575–4.925; the **gym** grows to 9.74 m (its west wall follows the office), the **suite court** to 8.39 m (it starts at the gym route, x 4.925). Suite, lobby, bath, dressing and bedroom unchanged.
- **V5.1 · gym:** an L-shaped gym (treadmill arm up to the suite bedroom, branch v5-gym-L) was studied and **rejected**: it took 3.45 m² from the suite court and put treadmill noise against the master bedroom. The gym stays a rectangle; the treadmill moves to its west end.
- **Clean rectangles (20 Oct, V5.0):** the office becomes one rectangle 4.25 × 4.75 m, the gym one straight bar 9.06 × 2.75 m (its L leg removed), the suite court runs the full width of the bedroom (7.71 × 2.0 m); office and gym doors are sliding, so no swing takes space.

Owner's office (the former freed roof) · 4.25 × 4.75 m, 20.19 m² · gym route · V5.1: 0.675 m west

- **Owner's decision (19 Oct 2026):** the freed south-west corner becomes the owner's office, an indoor room (walls, roof, windows) over the GF foyer / guest passage and the men's majlis; V5.1: its west 0.675 m is on the outward cantilever.
- **One rectangle** 4.25 × 4.75 m (the whole west end): sofa, coffee table and bookshelves in the north half; desk 1.6 × 0.8 m with two visitor chairs, a meeting corner and a cabinet in the south half. Windows west (two) and south (two).
- **Door** (sliding, lockable) from the gym route, ≈ 2 m from the end of the gallery. The gym route serves the office and the gym.
- **Bedroom 3** loses its small south window (that wall now faces the office); it keeps its west window.

Questions for the owner

zones, **anti-vibration pads** under its feet. Along the gym's depth (N–S) it would fill 2.5 of the 2.525 m clear depth and block the way to the rest of the gym, so it runs along the length.

- **Along the south wall to the east:** barbell + rack, kettlebells, pull-up bar (wall), pilates mat, one cardio machine (bike) and an **optional bench at the east end**. A 1.0 m clear walkway runs along the north wall from the door to the east end (checked: nothing in it, the door landing free).
- **Light:** the court stays 2.0 m deep in front of the clerestory windows; the gym's north wall is its south side, so the gym roof edge should end in a **railing, not a parapet** (solid top +7.20), to keep ≈ 80 % of the winter noon sun (V4.1 light check).

1. **Bedroom:** king bed facing the terrace as in V4.1, or another arrangement (TV wall, sitting area)?
2. **Terrace + light strip:** 16.77 m^2, 2.0 m deep, long and narrow: enough? Clerestory privacy (sill height or obscured glass) to settle.